1. hwcost calibration run `runs/final` (started 20261016T0459Z UTC)

Kind: benchmark / calibration run, descriptive report (observations only; the interpretation is in the study report `../../report.pdf`).
Study: `research/hwcost_model_20261016/`, protocol `PROTOCOL.md` frozen at commit `a9a457ec1b8ac619dfe789f5f24572a787ea806c` before this run.
Command: `./target/release/hwcost measure --out research/hwcost_model_20261016/runs/final`, then `hwcost fit --run … --fit-max-batch` 1024. Binary built from commit `a9a457ec1` (`cargo build --release --bin hwcost`). Started 20261016T0459Z, finished 20261016T0502Z (UTC); exit status 0.
Runs requested / completed: every arm below, 7 repetitions each, all completed; no arm failed or timed out.

1.1. Host and isolation

- CPU: Intel(R) Xeon(R) Processor @ 2.10GHz (GenuineIntel, signature 6/207/2), 4 logical CPUs, 1 NUMA node(s); virtualization: cpu_flag = hypervisor, systemd_detect_virt_vm = kvm.
- Caches (sysfs, as the guest sees them): L1 Data 48 KiB (CPUs 0), L2 Unified 2048 KiB (CPUs 0), L3 Unified 266240 KiB (CPUs 0-3).
- Binary compiled with target features: none of those probed (SSE4.2, AVX, AVX2, FMA, BMI1/2, ADX, AVX-512F/IFMA, PCLMULQDQ): the default x86-64 baseline. The CPU itself supports BMI2, ADX, AVX2 and AVX-512 (host capsule features); this build does not use them.
- Isolation level **L0**: shared cloud container, no core pinning, no frequency control. Arms were interleaved and their order rotated per repetition.
- Host capsule class id: `ECBENV2h2196cccd7a31`.

1.2. Correctness gates (before any timing)

For each field the batched walk (one batch inversion) was run 4 steps on 33 walks next to an unbatched reference (one inversion per walk); both agreed and every point satisfied $y^2 = x^3 + 7$ (stderr: `runs/final/stderr.txt`). Every timed kernel arm asserted zero degenerate steps.

1.3. Measured values

All values are medians over 7 repetitions (IQR where shown), each repetition about 100 ms.

Clock (M0). Dependent reg-reg `add` chain: 2.878 GHz; dependent `imul` chain (3-cycle latency): 2.797 GHz. Cycles below are ns × 2.878 GHz.

Field operations (M1), ns per operation (cycles in parentheses):

field	mul thr	mul lat	sqr thr	sqr lat	add thr	add lat	inv thr	inv lat
fp64	1.28 (4)	6.36 (18)	1.2 (3)	4.84 (14)	0.46 (1)	1.03 (3)	395 (1137)	376 (1081)
w1	2.21 (6)	8.46 (24)	1.74 (5)	6.59 (19)	0.54 (2)	1.4 (4)	629 (1810)	647 (1862)
w2	7.53 (22)	12.21 (35)	6.6 (19)	12.78 (37)	1.38 (4)	2.77 (8)	1360 (3914)	1355 (3898)
w3	15.07 (43)	21.19 (61)	13.43 (39)	19.91 (57)	2.28 (7)	3.81 (11)	2067 (5947)	2168 (6239)
w4	23.57 (68)	28.56 (82)	21.86 (63)	27.17 (78)	3.1 (9)	4.99 (14)	2976 (8564)	2969 (8545)

Memory read bandwidth (M2), GB/s, sequential read of the given size per thread:

threads	16 KiB	32 KiB	128 KiB	512 KiB	1 MiB	4 MiB	16 MiB	64 MiB	256 MiB	1024 MiB
1	71.3	67.1	68.7	71.1	69	29.7	22.4	20.4	8.8	9
4	269.4	247	214.5	212	238.1	98.8	86.5	43.9	34.7	35

Rho kernel (M3), ns per iteration on one thread, median and IQR, with the exact operation counts per iteration from the counted step (M5):

field	B	median ns	IQR	mul	mul (chain)	sqr / add	inv	model memory level
fp64	16	67.5	66.1–69.3	3	2	1 / 6	1/16	L1d
fp64	64	51.8	49.8–51.9	3	2	1 / 6	1/64	L1d
fp64	256	45.6	45.3–49.1	3	2	1 / 6	1/256	L1d
fp64	1024	48.7	46–50.3	3	2	1 / 6	1/1024	L1d
fp64	4096	47.1	45.1–47.6	3	2	1 / 6	1/4096	L2
fp64	16384	44.6	44.3–47.4	3	2	1 / 6	1/16384	L2
fp64	65536	45	44.4–46.8	3	2	1 / 6	1/65536	L3

fp64	262144	44.7	44.1–47.1	3	2	1 / 6	1/262144	L3
fp64	1048576	47.3	46.1–47.9	3	2	1 / 6	1/1048576	L3
fp64	4194304	49.3	47.4–50.8	3	2	1 / 6	1/4194304	L3
w1	16	68.6	65.4–69.7	3	2	1 / 6	1/16	L1d
w1	64	37.9	37.4–39.9	3	2	1 / 6	1/64	L1d
w1	256	32.3	32–33.5	3	2	1 / 6	1/256	L1d
w1	1024	28.4	27.8–29.7	3	2	1 / 6	1/1024	L1d
w1	4096	27.9	27.7–28.3	3	2	1 / 6	1/4096	L2
w1	16384	28.3	27.7–29	3	2	1 / 6	1/16384	L2
w1	65536	29.5	28.4–30.1	3	2	1 / 6	1/65536	L3
w1	262144	30.7	29.7–32.8	3	2	1 / 6	1/262144	L3
w1	1048576	30.6	30–31.6	3	2	1 / 6	1/1048576	L3
w1	4194304	30.6	30.2–30.7	3	2	1 / 6	1/4194304	L3
w2	16	153	148.3–156.7	3	2	1 / 6	1/16	L1d
w2	64	93	89.2–97	3	2	1 / 6	1/64	L1d
w2	256	83.1	79–84.1	3	2	1 / 6	1/256	L1d
w2	1024	74.8	73–78.5	3	2	1 / 6	1/1024	L2
w2	4096	76	74.3–77.2	3	2	1 / 6	1/4096	L2
w2	16384	73.8	71.8–76.2	3	2	1 / 6	1/16384	L2
w2	65536	75.7	72.1–76.8	3	2	1 / 6	1/65536	L3
w2	262144	75.7	72.9–76.9	3	2	1 / 6	1/262144	L3
w2	1048576	77	76.5–82	3	2	1 / 6	1/1048576	L3
w2	4194304	77.1	76.7–87.8	3	2	1 / 6	1/4194304	L3
w3	16	262.8	260.2–283.7	3	2	1 / 6	1/16	L1d
w3	64	163.7	161.1–165.1	3	2	1 / 6	1/64	L1d
w3	256	139.4	135.6–149.7	3	2	1 / 6	1/256	L1d
w3	1024	131.9	129.1–138.9	3	2	1 / 6	1/1024	L2
w3	4096	131.2	129.3–135.5	3	2	1 / 6	1/4096	L2
w3	16384	131.4	124.3–133.3	3	2	1 / 6	1/16384	L2
w3	65536	132.3	128.2–134.1	3	2	1 / 6	1/65536	L3
w3	262144	131.3	129.3–135.6	3	2	1 / 6	1/262144	L3
w3	1048576	138	136.9–139.4	3	2	1 / 6	1/1048576	L3
w3	4194304	135.4	133.6–140.4	3	2	1 / 6	1/4194304	DRAM
w4	16	384.8	381.2–431	3	2	1 / 6	1/16	L1d
w4	64	251.6	240.8–268.5	3	2	1 / 6	1/64	L1d
w4	256	207.8	200.1–211.1	3	2	1 / 6	1/256	L1d
w4	1024	196.2	193.6–200	3	2	1 / 6	1/1024	L2
w4	4096	193.4	189.1–199.1	3	2	1 / 6	1/4096	L2
w4	16384	190.6	188.9–194.1	3	2	1 / 6	1/16384	L3
w4	65536	195.3	188.1–200.5	3	2	1 / 6	1/65536	L3
w4	262144	195.1	194.2–211.1	3	2	1 / 6	1/262144	L3
w4	1048576	203.9	195.8–222.1	3	2	1 / 6	1/1048576	L3
w4	4194304	207.1	193.4–211.4	3	2	1 / 6	1/4194304	DRAM

Thread scaling (M4), B = 1024 walks per thread. The efficiency is ns(1 thread) / (T × aggregate ns per iteration at T threads):

field	T = 1	T = 2	T = 4
fp64	1	0.987	0.889
w1	1	0.92	0.903

w2	1	0.935	0.942
w3	1	1.007	0.939
w4	1	0.974	0.826

1.4. Graphs

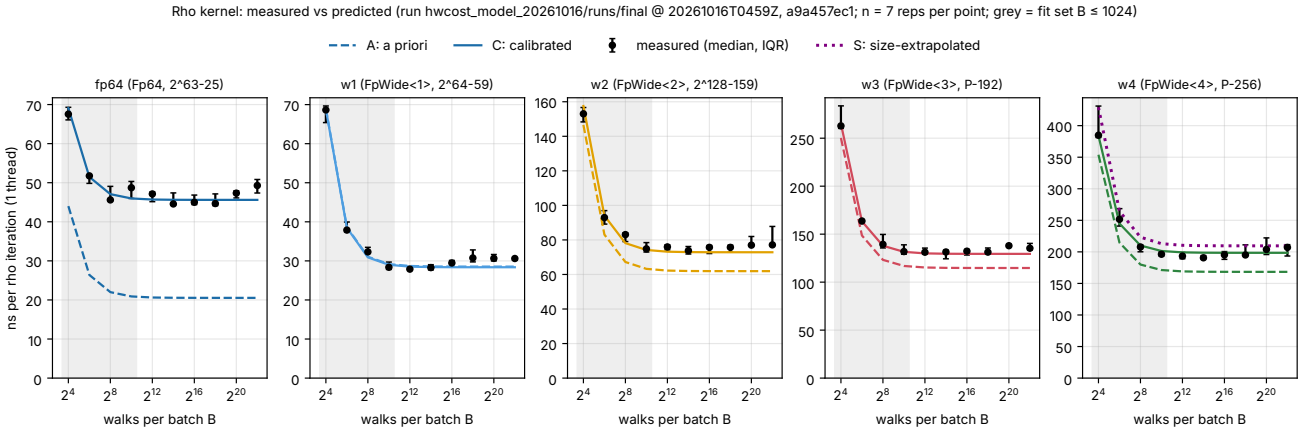


Figure 1: Measured ns per iteration (median, IQR; n = 7 per point) against batch size, with the protocol's model predictions A, C and S drawn over it. Grey: fit set. Run `runs/final`, 20261016T0459Z.

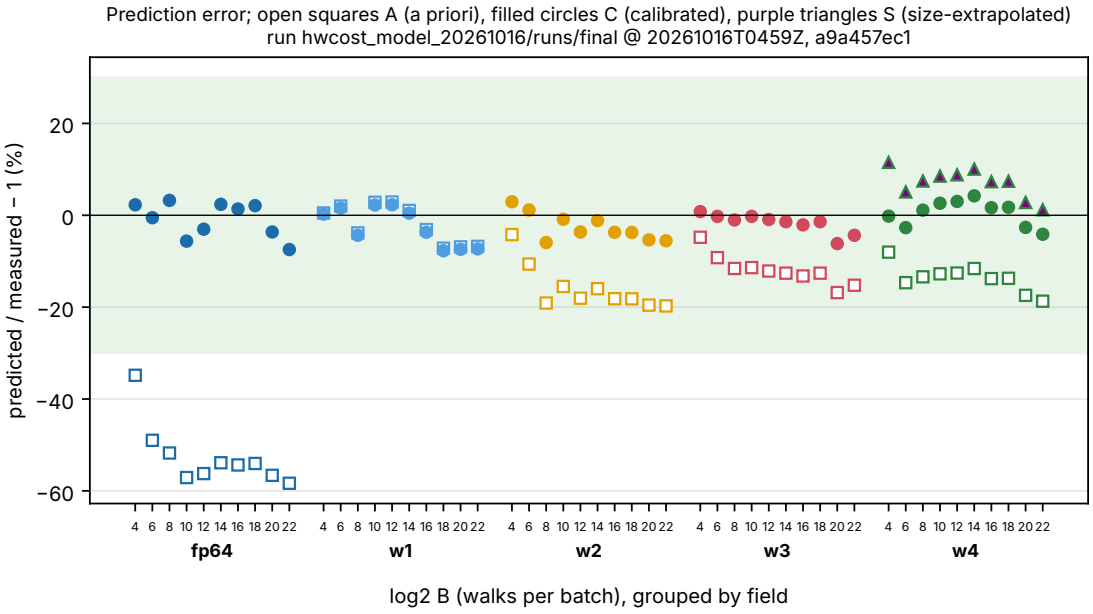


Figure 2: Prediction error `predicted / measured − 1` of models A, C and S per field and batch size; the band is the protocol's ±30% threshold.

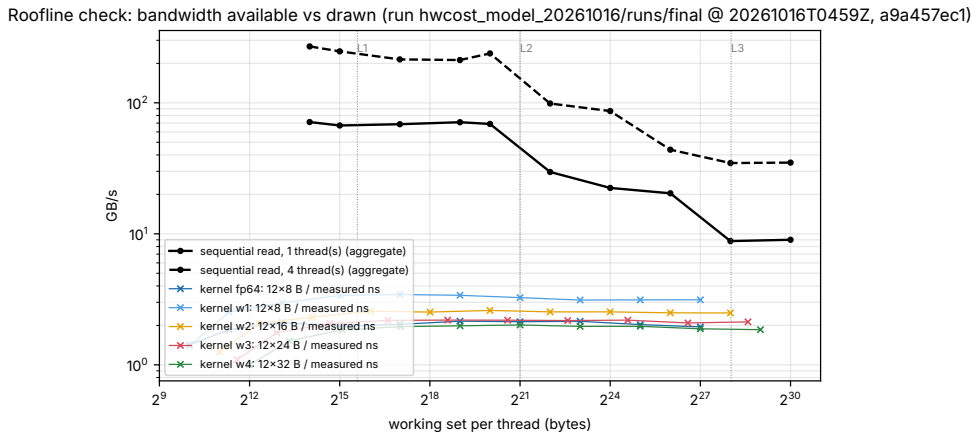


Figure 3: Measured sequential-read bandwidth (1 and 4 threads, aggregate) against size per thread, and the bandwidth each kernel arm drew (12 element-sized moves per iteration divided by its measured time).

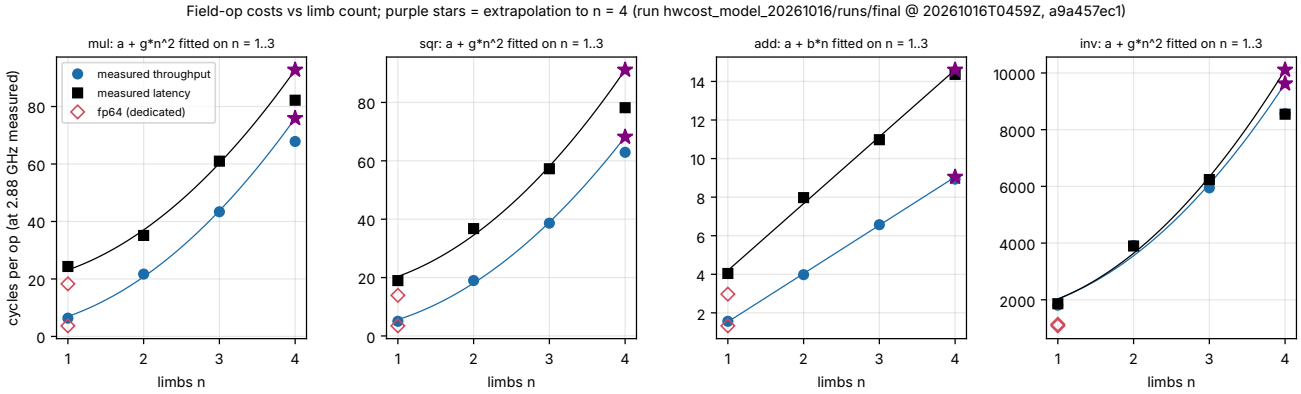


Figure 4: Measured cycles per field operation against limb count for `FpWide<n>`, with the size fits on n = 1..3 and their extrapolation to n = 4 (stars); `fp64` (dedicated backend) for reference.

1.5. Raw records

`runs/final/`: `measurements.json` (parameters, host capsule, caches, counts, every sample), `samples.csv` (one row per arm and repetition), `medians.csv`, `stderr.txt`, `commit.txt`, `started_utc.txt`, `finished_utc.txt`; fit outputs `profile.json`, `validation.json`, `validation.csv`, `fit.txt`. Graph source: `../../plot.py` (layout of the recorded values only).